STATISTICS

Chapter 5 Homework

Continuous models, areas, and waiting times

Sections 5.1-5.3 | Estimated time: about 1 hour

Complete all problems and show your work on separate paper. Plan for about 45-50 minutes of work and 10-15 minutes to check, correct, and submit.

Use available resources to check your answers and reasoning. Complete the self-evaluation in Blackboard, upload a clear photo of your one best representative page, and retain all your written work.

A four-function calculator is sufficient; needed advanced numerical values are supplied. Show probability setups, units, and brief explanations. Round probabilities to four decimals unless exact fractions are convenient. All scenarios and data are fictional.

Uniform waiting times

1. A shuttle wait X is modeled as uniform from 0 to 12 minutes. Sketch its density and find its height. Explain why the total area is 1.

2. For Problem 1 find P(3 < X < 9), P(X > 8), and P(X = 8). Mark each region on a sketch and distinguish a density height from a probability.

3. For Problem 1 find the mean and SD. Use the uniform SD formula (b - a)/√12 and √12 ≈ 3.464. Interpret the mean with units.

4. Find the 75th percentile of the shuttle wait. Explain which area lies to its left.

5. Given that the shuttle wait already exceeds 4 minutes, find P(X > 9 | X > 4). Compare it with P(X > 5). Is this uniform model memoryless?

Exponential waiting times

6. A service follows a constant-rate Poisson process, with mean time between arrivals 4 minutes. Let T be time until the next arrival. Identify its model, rate per minute, mean, SD, and possible values.

7. For Problem 6, use the tail formula below to find P(T > 2) and P(T ≤ 2). Use e to the power -0.5 ≈ 0.6065.

$$P(T > t) = e^{-t/4}, \qquad t \geq 0$$

8. For Problem 6 find P(2 < T < 6). Use e to the power -0.5 ≈ 0.6065 and e to the power -1.5 ≈ 0.2231. Explain why you subtract the two tails.

9. Find the median waiting time for Problem 6 using t = -4 ln(1 - p) for percentile p. Use ln(0.5) ≈ -0.6931. Why is this median less than the mean?

10. No arrival has occurred for 5 minutes. Under the same exponential model, find the probability of waiting more than 2 additional minutes. Explain how this differs from P(T > 7) measured from the original start.

11. Under the process in Problem 6, identify the distribution and mean for the number of arrivals in 20 minutes. Explain how the count variable differs from the waiting-time variable.

12. A model is continuous, but a recorded wait is rounded to the nearest minute. Explain why a displayed value of 8 can occur even though $P(X = 8) = 0$ for the exact continuous value.